

# Reporting Database

Basic Information, advantages, disadvantages:

[Reporting Database](#)

[Data Warehouse Reporting: Definition, Tips, Best Practices, and Reporting Tools | Databox.](#)

## Database Denormalization:

[Denormalization in Databases | GeeksforGeeks](#)

**why** → when join is expensive and queries need a lot of joins

**when** → normalization design is done as early as possible, but denormalization is done on demand basis

**Wrong updates can be done from different parts of the code**

Solution: Make sure update of one data is done by only one piece of code

**Updates can be slow.**

Solution: No big deal if update is not so frequent

When to build a Reporting database?

[When to build a separate reporting database? - Stack Overflow](#)

## How to sync reporting database

### Option 1: Scheduled update

- Daily, hourly or may be every 10 minutes
- Pro: Easy to manage
- Con: backdated data

### Option 2: Messaging

- Send message to reporting DB when there is an update in operation DB
- Pro: immediate update
- Con: difficult to manage

## Reporting database terminologies

**Measure/fact:** the data that we are interested about.

Example: selling price or sale.

**Fact** table: the table that contains the fact/measure.

Example: the sale table

**Dimension:** categories of fact.

Example: product

**Dimension** table: the table that contains the dimension.

Example: product table

[Schemas used in data warehouses: Star, Galaxy, and Snowflake](#)

[Star Schema vs Snowflake Schema – Difference Between Them](#)